When Predictive Weights Look Causal: An Audit Protocol for Interpretability-Based Discovery

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Abstract

Readable model weights are tempting evidence about the mechanisms that generated scientific data. We audit one such candidate: the endpoint-product score of a next-step time-series predictor. A sequence of initially plausible architecture, benchmark, intervention, and nonlinearity stories does not survive controls for direction conventions, lag horizons, label provenance, data budgets, intervention targets, and classical baselines. Correcting a source/target transpose and matching lag horizons changes Lorenz-96 PCMCI AUROC from 0.986 in our earlier preprint to 0.839. On target-aware, size-matched random forcing, all four tested methods improve, so the effect is not bottleneck-specific. In a fixed nonlinearity confirmation, the only gate-passing bottleneck advantage is identical at every nonlinearity strength, vanishes when the sample size doubles, and arises because the prediction-tuned Lasso comparator returns all-zero cross-variable scores. We demonstrate a reproducible falsification protocol on one endpoint-product score family, with reporting and responsible-use boundaries; transfer to other interpretability signals remains untested. This negative study illustrates how a readable predictive representation can remain useful for hypothesis generation without constituting causal knowledge.

1 Introduction

Interpretability can expose regularities that domain experts did not encode explicitly, making it a promising interface between learned representations and scientific discovery. But an explanation can be readable without identifying the data-generating mechanism. In time series, predictive association, Granger predictability, intervention response, and structural causation answer different questions [Granger, 1969, Pearl, 2009, Peters et al., 2017]. Confusing them can turn a useful model diagnostic into an unsupported scientific claim.

Our motivating case is a next-step predictor with input and output projections. For a non-lagged model, the candidate score is

$$S = |W_{\text{out}} W_{\text{in}}|, \qquad S_{ij} \text{ read as the strength of } j \to i. \tag{1}$$

The score is cheap, compact, and often graph-like. It arose while studying a Mamba-style state-space forecaster [Gu and Dao, 2024], but a readable pattern does not establish that the architecture discovered causal structure. Our earlier preprint found that a plain linear bottleneck matches or exceeds the state-space readout [Lade et al., 2026], so the corrected experiments here audit the linear factorized predictor directly.

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We make three contributions. First, we organize implementation, estimand, and provenance checks into an audit protocol for discovery claims. Second, we apply it to corrected observational, interventional, and controlled nonlinear experiments while retaining null and contradictory outcomes. Third, we pair the empirical audit with a responsible-use and reproducibility boundary. We do not propose Eq. 1 as a causal-discovery method. The checks are reusable recommendations; their empirical validation here is restricted to one predictive score family, not interpretability methods in general.

Relation to prior work. Time-series work spans Granger/VAR methods, conditional-independence tests, and neural estimators [Runge et al., 2019, Runge, 2020, Tank et al., 2022, Nauta et al., 2019, Pamfil et al., 2020]. Interpretable neural Granger models explicitly design the predictor around a causal estimand [Marcinkevičs and Vogt, 2021]; causal representation learning likewise emphasizes assumptions connecting representations to interventions [Schölkopf et al., 2021]. Our question is complementary: what should be checked *before* treating an ordinary predictive weight as scientific evidence? Attention studies question whether readable weights faithfully explain predictions [Jain and Wallace, 2019], with important qualifications about the explanatory claim and evaluation [Wiegreffe and Pinter, 2019]. Saliency sanity checks show that visually plausible maps can fail model- and data-randomization tests [Adebayo et al., 2018]; Rudin [2019] distinguishes inherently interpretable models from post-hoc explanations. Our novelty is not the warning that readability is insufficient. It is a worked audit of the additional inference from predictive weights to causal structure, using orientation, lag, provenance, matched-budget, and intervention-target checks.

2 Candidate and Audit Design

Lag-resolved candidate. Most corrected experiments use a factorized predictor of x_t from $[x_{t-1}, \ldots, x_{t-p}]$. Splitting the input projection by lag gives

$$S_\tau = |W_{\text{out}} W_{\text{in},\tau}|\,, \quad \tau = 1, \ldots, p,$$

with convention $S_\tau[\text{target}, \text{source}]$. We zero self-edges. The predictor is linear: W_{in} maps the Kp lagged inputs to width d and W_{out} maps back to K outputs, with no activation. We use $d = 16$, except $d = 32$ in Section 3.3; because d can exceed K, we call it factorized rather than rank-limited. Table 1 aggregates lags by their maximum score; Section 3.3 instead evaluates the flattened off-diagonal lag tensor, retaining lag identity. The "Lagged-BN" and "Linear-BN" labels (BN: bottleneck, our historical name for this factorized predictor) refer to this prediction-trained model, not to an intervention-aware causal model. Lasso and Ridge use the same lagged rows and choose regularization by prediction validation rather than graph labels.

Representation and protocol checks. We test direction, lag, and diagonal conventions on planted edges. Package adapters are explicit: Tigramite stores values as $[\text{source}, \text{target}, \text{lag}]$, whereas our score uses $[\text{target}, \text{source}]$. Strictly lagged evaluation excludes contemporaneous matrices. The original controlled comparisons match usable observations, lag horizon, and preprocessing. The added observational baselines share input arrays and a maximum lag horizon; VAR-LiNGAM selects its lag order within that horizon (Appendix B). Independent trajectories remain separate so no lag window crosses a boundary.

Intervention-aware estimand. Each intervention segment records its target. When variable j is externally forced, the outcome equation for j is excluded from that segment for every method; other outcome equations remain usable. A combined observational/interventional arm is compared with an observational arm having the same number of usable lagged rows. Constant clamps, additive soft forcing, and per-step random forcing are separate conditions.

Provenance and inference. Every graph is tagged as exact simulated, literature-derived soft, or heuristic soft. Independent data-generating seeds are the replication unit; model-seed spread on fixed data is only descriptive. In the original intervention and nonlinearity studies, paired effects receive bootstrap 95% intervals and one-sided Wilcoxon tests with Holm correction over each declared family. Hyperparameters and gates are fixed before graph evaluation. The post-review trajectory comparison reports exploratory bootstrap intervals without significance tests.

Table 1: Corrected AUROC at lag horizon 3. Climate and finance have soft labels and are sensitivity analyses, not causal leaderboards. Lagged-BN reports mean ± standard deviation over three model initializations on fixed data; other rows are deterministic fits on those same data. This is a descriptive comparison, not an estimate of across-dataset uncertainty. Finance Lasso selects the largest penalty in its grid and returns all-zero cross-variable scores, so its 0.500 is a tie, not a ranking. Ridge also selects a grid-boundary penalty on climate (smallest) and finance (largest) but returns untied scores.

Method	Climate	Finance	Lorenz-96
Lasso	0.617	0.500	0.973
Ridge	0.617	0.597	0.973
Lagged-BN (3 seeds)	0.626 ± 0.021	0.596 ± 0.051	0.881 ± 0.018
PCMCI	0.407	0.609	0.839
PCMCI+	0.605	0.579	0.875
Bivariate Granger	0.605	0.512	0.989

3 Corrected Evidence

3.1 Benchmark provenance and orientation

Table 1 separates two observational sensitivity analyses from an exact simulated graph. Climate contains six NOAA indices ($T = 757$) and three literature-derived teleconnection labels [Trenberth and Hurrell, 1994, Newman et al., 2016, Delworth and Mann, 2000]; the definitionally coupled ENSO/SOI pair is excluded. Finance contains ten sector ETF return series ($T = 1893$) and six heuristic labels. Lorenz-96 is a $K = 10, T = 1500$ simulation with the exact 30-edge interaction graph [Lorenz, 1996].

Our earlier preprint [Lade et al., 2026] reported a Lorenz-96 PCMCI value of 0.986 from a transposed score matrix at lag horizon 5; at the correct orientation and the shared horizon 3 it is 0.839. The preprint's DYNOTEARS and VAR-LiNGAM values also included contemporaneous coefficients (Appendix B). Its benchmark values are superseded by Table 1 and Appendix B. On the exact graph, Granger and tuned linear regression have higher observed scores than PCMCI/PCMCI+ in this configuration. The Lagged-BN score still contains readable dependency signal, but its 0.881 ± 0.018 remains below Granger/Lasso/Ridge ($0.989/0.973/0.973$). Its spread over model seeds 42–44 is initialization stability, not independent uncertainty. Climate and finance cannot support a causal ranking. Near-ceiling linear and Granger scores make this an implementation and interpretation check, not a demanding test of general causal-discovery performance. The exact reference graph describes the continuous-time generating equations; subsampling need not preserve its interpretation as a discrete-time conditional-independence graph.

Post-review trajectory replication. We additionally evaluate ten independently initialized Lorenz-96 trajectories (seeds 81000–81009), retaining the original $K = 10, T = 1500$, forcing, sampling, lag horizon, preprocessing, and estimator settings. Each trajectory is the replication unit; the bottleneck averages model seeds 42–44 within it. Table 2 reports the resulting uncertainty for all methods. Lasso, Ridge, Granger, and VAR-LiNGAM exceed Lagged-BN with paired intervals excluding zero; the PCMCI and PCMCI+ intervals include zero; and DYNOTEARS is lower, with an interval excluding zero. On AUPRC, Lagged-BN is the lowest of all eight methods. These exploratory follow-ups extend replication, not graph-family coverage. DYNOTEARS and VAR-LiNGAM are run on the identical exported arrays in an isolated compatible environment; neither is evaluated in the intervention experiments. Package defaults and assumptions are detailed in Appendix B.

3.2 Intervention semantics and method specificity

Both intervention audits use random stable linear-Gaussian VAR(3) systems with $K \in \{10, 20\}$, $T = 300$ observational rows, and 10 independent seeds: cross-edge probability $.2/3$ per ordered pair and lag, coefficients of magnitude $U(.1, .25)$ with random sign, self-coefficients $U(.2, .4)$ at each lag, rescaled when needed to companion spectral radius $.85$, and noise standard deviation $.1$. Each variable is then forced for 50 steps. For method m, the intervention gain is

$$\Delta_m = \mathrm{AUROC}_m(\text{combined}) - \mathrm{AUROC}_m(\text{size-matched observational}).$$

Table 2: Post-review Lorenz-96 replication over ten independent trajectories. AUROC and AUPRC are means with 95% percentile bootstrap intervals over trajectories; bottleneck initializations are averaged within each trajectory. The final column is the paired AUROC difference, method minus Lagged-BN, within each trajectory (positive favors the listed method), with its own trajectory-bootstrap interval. All results are exploratory.

Method	AUROC [95% CI]	AUPRC [95% CI]	Δ AUROC [95% CI]
Lagged-BN	0.872 [0.854, 0.890]	0.765 [0.731, 0.799]	—
Lasso	0.955 [0.942, 0.968]	0.933 [0.916, 0.949]	0.083 [0.064, 0.100]
Ridge	0.954 [0.941, 0.967]	0.931 [0.914, 0.948]	0.082 [0.063, 0.100]
PCMCI	0.855 [0.841, 0.870]	0.831 [0.820, 0.844]	−0.017 [−0.043, 0.008]
PCMCI+	0.844 [0.800, 0.885]	0.836 [0.798, 0.874]	−0.028 [−0.065, 0.007]
Granger	0.977 [0.963, 0.988]	0.964 [0.944, 0.980]	0.105 [0.094, 0.116]
DYNOTEARS	0.800 [0.783, 0.816]	0.851 [0.833, 0.866]	−0.073 [−0.094, −0.047]
VAR-LiNGAM	0.900 [0.880, 0.917]	0.845 [0.818, 0.868]	0.027 [0.013, 0.042]

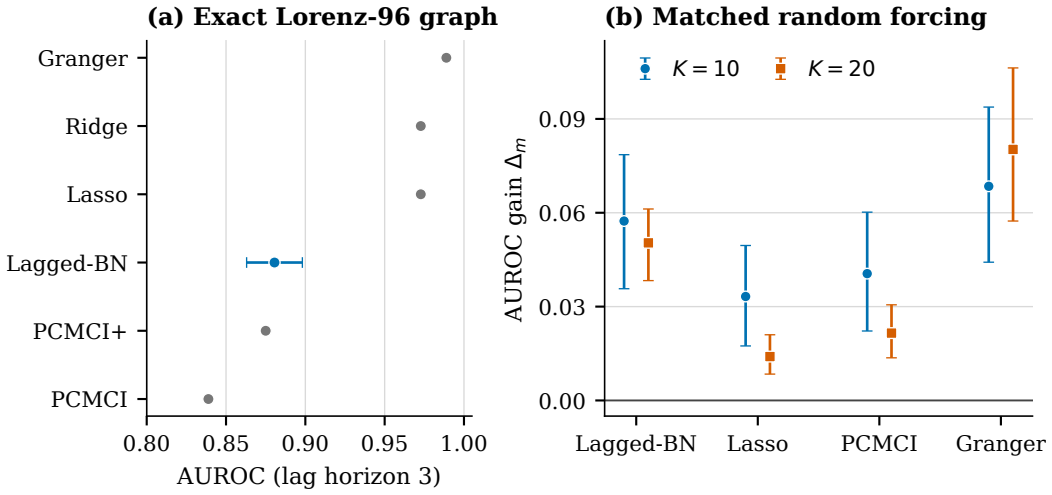


Figure 1: Corrected evidence summary. (a) Lorenz-96 AUROC against the exact graph; Lagged-BN reports mean ± standard deviation over three model initializations, while the baselines are deterministic. (b) Target-aware random-forcing gain relative to a size-matched observational arm, with bootstrap 95% intervals over 10 independent data seeds, $T = 300$.

The semantics audit tests whether $\Delta_{\mathrm{BN}} - \Delta_m > 0$ for Lasso and Ridge, using lag-resolved AUROC. Holm-significant positive differences are only 0.012–0.021 AUROC. Constant clamps yield no positive bottleneck advantage; random and soft forcing produce selected small differences, not a general mechanism.

The four-method audit then applies target-aware random forcing to the bottleneck, Lasso, PCMCI, and Granger, using lag-aggregated AUROC. Figure 1 shows positive gains for every method at both dimensions; Appendix C lists the values. Combined-arm AUROC is at least .99 for every method, so each Δ_m is bounded by that method's observational shortfall; between-method differences in gain, including the Granger contrast below, partly reflect weaker observational recovery.

All eight within-method tests reject after Holm correction ($p_{\mathrm{Holm}} = 0.0078$), so the gain is not bottleneck-specific. An exploratory paired Granger-minus-Lagged-BN contrast does not reject at $K = 10$ ($p_{\mathrm{Holm}} = 0.2783$) and is positive at $K = 20$ ($+0.030$, $p_{\mathrm{Holm}} = 0.0273$). The evidence therefore supports neither bottleneck specificity nor method invariance. Target-aware Lasso also improves, so masking forced outcomes does not remove its gain in these settings. Positive gains across four methods do not imply equal gains or benefits for untested methods.

3.3 Nonlinearity confirmation

The controlled generator varies only the planted edge function

$$g_s(x) = a(s)\big[(1-s)x + s\sigma\tanh(x/\sigma)\big],$$

where $\sigma \approx .085$ is the stationary source standard deviation of a fixed Gaussian AR(1) reference and $a(s)$ equalizes marginal scale under it. There are only two planted cross-variable edges, at lags 3 and 4; self-dynamics, edge coefficients, noise, lag structure, and seeds remain fixed. This is a sparse controlled check, not a broad nonlinear benchmark. It tests our earlier preprint's exploratory finding of a bottleneck advantage under mild nonlinearity at $K = 20, T = 300$ [Lade et al., 2026]. We compare a fixed Linear-BN ($d = 32$, sparsity coefficient 10^{-3}) with prediction-tuned Lasso over $K \in \{10, 20\}$, $T \in \{300, 600\}$, $s \in \{0, .2, .3, .4, .6, 1\}$, and 10 data seeds. The primary metric is lag-resolved AUROC at horizon 5. A cell is one (K, T, s) setting with ten paired data seeds. It passes only if all six conditions hold: mean AUROC difference $\geq .03$; paired bootstrap 95% lower bound > 0; one-sided Wilcoxon $p_{\text{Holm}} < .05$ across all 24 cells; mean bottleneck AUPRC greater than Lasso's; mean bottleneck AUPRC greater than edge prevalence; and mean held-out prediction MSE no greater than 1.10 times Lasso's. A regime requires the *same* two or more $s \in \{.2, .3, .4\}$ to pass at both dimensions at one common T. Appendix A gives splits, metric definitions, and all cell results.

No T satisfies that rule. Six cells pass, all at $K = 20, T = 300$, so the passing cells reflect one (K, T) setting rather than any nonlinearity strength. At $K = 20, T = 300$, Linear-BN minus Lasso AUROC is $+0.50$ for *every* s, including 0 and 1. At $K = 10, T = 300$, it is $+0.05$ for every s, but no cell passes: each has bootstrap lower bound zero and adjusted $p = 1$. At $T = 600$, both methods reach AUROC 1.00 throughout.

A post-review diagnostic explains these differences. Linear-BN attains AUROC and AUPRC 1.00 in all 240 fits, so every nonzero difference comes from the comparator. Re-fitting reproduces the Lasso penalty, AUROC, and test MSE for all 240 saved fits. At $K = 20, T = 300$, all 60 fits select $\alpha = 1$, the largest value in the grid, and have zero off-diagonal lag scores; at $T = 600$, all 120 fits select $\alpha = .1$ and attain perfect AUROC. The $+0.50$ difference is thus against an uninformative edge ranking chosen by prediction validation: comparator over-regularization at $K = 20, T = 300$, not a nonlinearity-specific advantage. The design cannot resolve a nonlinearity effect in either direction, and the gates lacked a check that the comparator's ranking is non-degenerate; Section 4 adds that check. No graph-label-based retuning replaces the original comparison.

4 Implications, Limitations, and Responsible Use

A practical falsification checklist. Before elevating an interpretation to a discovery claim: (i) test direction, lag, and diagonal conventions on planted examples; (ii) match data budgets, preprocessing, boundaries, and seeds; (iii) encode intervention targets and state intervention semantics; (iv) compare against sparse and classical baselines under the same estimand, verify that each comparator gives a non-degenerate ranking (no all-tied scores), and report any penalty selected at a grid boundary; (v) distinguish exact graphs from soft labels; and (vi) fix variants before evaluation, report multiplicity-adjusted effects, and retain failures.

Limitations. The evidence covers one score family, two dimensions ($K = 10, 20$) in the intervention and nonlinearity studies, one intervention sample size ($T = 300$), and two nonlinearity sample sizes ($T = 300, 600$). Lorenz-96 provides one exact dynamical graph family; the controlled generator has only two planted cross-variable edges. Climate and finance have sparse, incomplete soft labels. Trajectory replication improves uncertainty reporting but does not make Lorenz-96 harder or validate another interpretability signal. DYNOTEARS and VAR-LiNGAM extend the observational comparison only; their linear-model assumptions need not hold for Lorenz-96. Broader graph, confounding, lag, noise, and intervention coverage remains open.

Responsible use and reproducibility. These scores must not prescribe scientific interventions, climate policy, financial trades, or other consequential decisions. Climate and finance labels are incomplete hypotheses, and Lorenz-96 is simulated. Predictive dependence can guide questions, but causal use requires domain review, explicit assumptions, validated interventions, and evidence outside the interpreted model. Code and result artifacts are available at `https://github.com/`

`ankitlade12/predictive-weights-audit`. The repository's release archive contains the official template, pinned environments, tests, commands, input hashes, submitted and post-review result artifacts, and mechanically generated table values. New analyses are explicitly distinguished from the original fixed-gate experiments.

What the checklist establishes. The checklist diagnoses reasons an apparent discovery can be unsupported. Orientation fixtures test an implementation convention; they do not identify causal structure. Matching data budgets removes a sample-count confound; it does not equate information content. Target masking respects which equations are externally changed; it does not guarantee intervention identification. Classical comparisons test specificity to the interpreted model; they do not make every baseline causally valid. Non-degeneracy checks show that a comparison is informative; they do not make the comparator optimal. Exact simulation labels and soft observational hypotheses require different interpretations. Each check targets a specific failure mode: failing one undermines a particular discovery claim, but passing all of them does not establish causal identification. Validating the checklist on another interpretation signal, harder graphs, and broader confounding or intervention settings remains future work.

5 Conclusion

The endpoint bottleneck is informative precisely because its causal story looked plausible. Orientation, provenance, matched estimands, intervention semantics, and classical baselines remove the strong interpretation. The remaining contribution is a worked falsification protocol for one predictive score family, with checks that can be tested on further interpretation methods before their outputs become scientific claims.

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A Nonlinearity decision rule and complete cell results

Data and splits. The controlled generator has $K \in \{10, 20\}$ variables, $T \in \{300, 600\}$ retained observations, 300 burn-in steps, self coefficient .35, edge coefficient .55, and innovation standard deviation .08. Its only cross-variable edges are $0 \to 1$ at lag 3 and $2 \to 3$ at lag 4; these are unchanged across strengths. The nonlinearity is scaled using 64-point Gauss–Hermite quadrature under a fixed Gaussian AR(1) reference, independently of experimental data and seeds. This controls reference marginal scale; it does not guarantee equality of every finite-sample distribution.

Each raw series is split chronologically into 80% training and 20% test observations. Per-variable means and scales are fitted on the training split only. Lag windows are constructed separately within each split. Lasso uses the final 25% of the training series for prediction-validation selection from $\{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1\}$, then refits on the entire training series. This is the existing multi-output `MultiTaskLasso` baseline; its penalty couples coefficients across outcome variables. "Lasso" in the paper refers to this implementation when outcome masks are absent. With intervention target masks, regressions are fitted per outcome. No graph labels select the penalty. The linear bottleneck uses width 32, 80 epochs, batch size 64, learning rate .001, weight decay .0001, and endpoint sparsity coefficient .001. Width 32 exceeds K here, so a strict output-rank bottleneck is not imposed in this experiment; factorization and optimization/regularization effects must not be conflated with rank restriction.

Metrics and inference. The primary edge universe is the $5K(K-1)$ off-diagonal lag–target–source entries, with two positive labels. Edge prevalence is therefore $2/[5K(K-1)]$: .004444 for $K = 10$ and .001053 for $K = 20$. AUROC and AUPRC are computed on that universe; AUPRC uses trapezoidal integration of the precision–recall curve, not average precision. For tied scores its interpolation can differ from a prevalence-valued random-ranking reference; the prevalence gate is a heuristic floor, not a calibrated null test. Prediction MSE averages squared errors over held-out lagged rows and variables in training-standardized units.

Data seeds 60000–60009 are paired between methods and reused across cells; model seeds 70000–70009 accompany them. Reuse across strengths does not create additional independent replications. For each cell, the effect is the mean of ten paired AUROC differences. The 95% percentile interval resamples those ten pairs 10,000 times. One-sided Wilcoxon tests use the alternative of positive bottleneck-minus-Lasso differences; all-zero differences are marked unevaluable and contribute a conservative placeholder $p = 1$ to the correction family. They cannot pass the significance gate. Holm adjustment is across the 24 cell tests. The AUPRC and MSE gates compare cell means and are descriptive thresholds, not additional significance tests. Neither the original gates nor their multiplicity family was changed after review.

Writing $\overline{A}_B, \overline{A}_L$ for mean AUPRC and $\overline{E}_B, \overline{E}_L$ for mean test MSE, a positive cell requires

$$\overline{\Delta} \geq .03, \quad L_{95} > 0, \quad p_{\text{Holm}} < .05, \quad \overline{A}_B > \overline{A}_L, \quad \overline{A}_B > \pi_K, \quad \overline{E}_B \leq 1.10\overline{E}_L.$$

The regime rule is

$$\exists T: \quad |\{s \in \{.2, .3, .4\} : \forall K \in \{10, 20\}, \ \text{pass}(K, T, s)\}| \geq 2.$$

Table 3 gives every cell, including the linear and fully nonlinear controls. Six positive cells confined to $K = 20, T = 300$ do not meet this rule.

Post-review diagnostic. Re-fitting the original Lasso configurations reproduces selected penalties, AUROC, and held-out MSE for all 240 runs. The diagnostic counts nonzero off-diagonal lag scores, without inferring whether self coefficients are zero. All 60 $K = 20, T = 300$ fits select $\alpha = 1$ and have zero off-diagonal scores. At $K = 10, T = 300$, six fits (one data seed across all six strengths) select $\alpha = 1$ and have zero scores; the other 54 select .1. All 120 $T = 600$ fits select .1. These observations locate the uninformative ranking in the prediction-selected comparator; they do not establish a universal regularization mechanism or justify selecting penalties by graph accuracy.

B Post-review replication and additional baselines

Scope and pairing. The follow-up was specified after receiving reviewer feedback and is exploratory. The ten Lorenz-96 trajectories share the original graph and settings ($K = 10, T =$

Table 3: All original nonlinearity cells (ten paired seeds each). Δ and CI refer to lag-resolved AUROC. PR reports bottleneck/Lasso mean AUPRC; MSE reports their mean-MSE ratio. Failure labels are effect ($\Delta < .03$), CI (lower bound ≤ 0), p (adjusted $p \geq .05$), and PR (no AUPRC superiority). A dash denotes an unevaluable all-zero Wilcoxon contrast. No cell fails the prevalence or MSE gate. A Lasso AUPRC near .50 is the trapezoidal value for all-tied zero scores.

K	T	s	Δ	95% CI	Holm p	PR B/L	MSE B/L	Result
10	300	0	0.05	[0.00,0.15]	1.0000	1.000/0.950	1.073	CI,p
10	300	0.2	0.05	[0.00,0.15]	1.0000	1.000/0.950	1.073	CI,p
10	300	0.3	0.05	[0.00,0.15]	1.0000	1.000/0.950	1.073	CI,p
10	300	0.4	0.05	[0.00,0.15]	1.0000	1.000/0.950	1.073	CI,p
10	300	0.6	0.05	[0.00,0.15]	1.0000	1.000/0.950	1.073	CI,p
10	300	1	0.05	[0.00,0.15]	1.0000	1.000/0.950	1.073	CI,p
10	600	0	0.00	[0.00,0.00]	—	1.000/1.000	1.049	effect,CI,p,PR
10	600	0.2	0.00	[0.00,0.00]	—	1.000/1.000	1.049	effect,CI,p,PR
10	600	0.3	0.00	[0.00,0.00]	—	1.000/1.000	1.049	effect,CI,p,PR
10	600	0.4	0.00	[0.00,0.00]	—	1.000/1.000	1.049	effect,CI,p,PR
10	600	0.6	0.00	[0.00,0.00]	—	1.000/1.000	1.048	effect,CI,p,PR
10	600	1	0.00	[0.00,0.00]	—	1.000/1.000	1.048	effect,CI,p,PR
20	300	0	0.50	[0.50,0.50]	0.0234	1.000/0.501	1.068	pass
20	300	0.2	0.50	[0.50,0.50]	0.0234	1.000/0.501	1.068	pass
20	300	0.3	0.50	[0.50,0.50]	0.0234	1.000/0.501	1.068	pass
20	300	0.4	0.50	[0.50,0.50]	0.0234	1.000/0.501	1.068	pass
20	300	0.6	0.50	[0.50,0.50]	0.0234	1.000/0.501	1.068	pass
20	300	1	0.50	[0.50,0.50]	0.0234	1.000/0.501	1.069	pass
20	600	0	0.00	[0.00,0.00]	—	1.000/1.000	1.052	effect,CI,p,PR
20	600	0.2	0.00	[0.00,0.00]	—	1.000/1.000	1.052	effect,CI,p,PR
20	600	0.3	0.00	[0.00,0.00]	—	1.000/1.000	1.052	effect,CI,p,PR
20	600	0.4	0.00	[0.00,0.00]	—	1.000/1.000	1.052	effect,CI,p,PR
20	600	0.6	0.00	[0.00,0.00]	—	1.000/1.000	1.052	effect,CI,p,PR
20	600	1	0.00	[0.00,0.00]	—	1.000/1.000	1.052	effect,CI,p,PR

$1500, F = 10$, integration step .01, subsampling factor 10, 500 discarded integration steps), but use independently drawn initial states from seeds 81000–81009. They do not vary graph topology, forcing, noise, or sampling interval. Full-series standardization matches the submitted graph-recovery benchmark; these are not held-out forecasting results. Each estimator receives the same exported standardized array, verified by SHA-256 across environments, with lag horizon 3. All original estimator settings are unchanged. The bottleneck's three initialization scores are averaged within each trajectory, leaving ten units for inference.

Means and 95% percentile bootstrap intervals in Table 2 use 10,000 trajectory resamples with seed 82000. Paired differences are formed within each trajectory before resampling. Intervals are marginal, not simultaneous; no multiplicity-adjusted superiority claim is made. We retain every planned trajectory and report package warnings and failures in the machine-readable artifacts. A failed or missing method/trajectory prevents generation of the complete comparison table.

Additional baseline configuration. The optional baselines use an isolated Python 3.10.19 environment with CausalNex 0.12.1 and LiNGAM 1.11.0; its exact installed package versions are included in the archive. DYNOTEARS [Pamfil et al., 2020] uses $p = 3$, $\lambda_W = \lambda_A = .1$, maximum 100 iterations, acyclicity tolerance 10^{-8}, and weight threshold zero. VAR-LiNGAM [Hyvärinen et al., 2010] uses maximum lag 3, BIC lag selection within that horizon, pruning enabled, and random state zero. Their package-default settings are not graph-label tuned. This comparison therefore has a common maximum lag horizon, not necessarily the same selected lag order or tuning budget. Contemporaneous coefficients are excluded from scoring; only positive-lag, off-diagonal coefficients are aggregated by maximum absolute value in $[\mathrm{target}, \mathrm{source}]$ orientation.

On the original seed-0 Lorenz-96 array, the added baselines give AUROC 0.833 (DYNOTEARS) and 0.896 (VAR-LiNGAM). They were not part of the submitted Table 1; that historical comparison is preserved. Our earlier preprint [Lade et al., 2026] reported higher Lorenz-96 values (0.983 and 0.968) from adapter versions that also aggregated contemporaneous coefficients; the strictly lagged values here supersede them. Neither method is run on the intervention or nonlinearity conditions in this revision. Their linear structural-model assumptions (including non-Gaussian independent

innovations for VAR-LiNGAM) are not established for the subsampled Lorenz-96 trajectories. Their scores are therefore bounded empirical comparisons, not proofs of causal validity.

C Intervention gains

Table 4 lists the values plotted in Figure 1(b).

Table 4: Target-aware random-forcing gain Δ_m (lag-aggregated AUROC): paired mean and bootstrap 95% interval over 10 independent data seeds, $T = 300$.

Method	$K = 10$	$K = 20$
Lagged-BN	$0.057\ [0.036, 0.079]$	$0.050\ [0.038, 0.061]$
Lasso	$0.033\ [0.017, 0.050]$	$0.014\ [0.008, 0.021]$
PCMCI	$0.041\ [0.022, 0.060]$	$0.022\ [0.014, 0.031]$
Granger	$0.068\ [0.044, 0.094]$	$0.080\ [0.057, 0.106]$